



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

CISCO NETWORKING ACADEMY · VERIFIED INTERNSHIP

Network Technician Career Path

An Internship Presentation Report

INTERNSHIP DURATION

April 2026 – June 2026

STUDENT CREDENTIALS

Ankit Kumar

Enrollment / Roll No.

2410031474

Program & Batch

B.Tech (CSE)

HOST ORGANIZATION

Cisco Networking Academy

Delivered through IILM University

Candidate & Program Overview



Candidate Details

Student Name: ANKIT KUMAR

Roll Number: 2410031474

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2024-2028

University: IILM University, Greater Noida, U.P.



Internship Profile

Program Name: Cisco Networking Academy Verified Path

Specialization Domain: Network Technician / Computer Networking

Duration: April 2026 – June 2026

Issued On: June 16, 2026

Delivered Via: School of Computer Science & Engineering, IILM

Organization Profile: Cisco Networking Academy

Cisco Networking Academy is Cisco Systems' global IT skills and career-building program, delivered in partnership with academic institutions worldwide to build industry-ready networking talent.



IILM Collaboration: Delivered through the School of Computer Science and Engineering, IILM University, empowering student networking talent.



Industry Alignment: Follows Cisco's globally recognized curriculum, bridging academic learning and real-world network-technician skills.



Core Focus Domains: Networking Fundamentals, Ethernet Switching, IP Addressing, Wireless Configuration, and Network Security.



Program Snapshot

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Courses completed

100%

Verified credential

Jun 2026

Issued & verified

Problem Statement & Program Context



The Skills Gap

While engineering curricula extensively cover networking theory, students often lack hands-on exposure to configuring, securing, and troubleshooting real network devices.

Understanding protocols and addressing on paper is only one part of network literacy; diagnosing connectivity issues and securing access is rarely practiced.



The Program's Scope

This internship addresses the full scope of entry-level network technician tasks — standards, switching, addressing, wireless, routing, and troubleshooting.

Mastering Cisco Packet Tracer simulation to transform classroom concepts into configured, secured, and diagnosable network environments.

Core Project Objectives



Standards & Protocols

Explain the importance of standards and protocols in network communications.



Ethernet & Switching

Explain how communication and Ethernet operate within a switched network.



Wireless Configuration

Configure an integrated wireless router and client to connect securely to the internet.



Secure Access

Configure secure user access and apply effective troubleshooting methodology.

Technologies & Networking Toolkit



Simulation & Design

Environment: Cisco Packet Tracer

Design: Topology building & testing



Addressing & Routing

Schemes: IPv4 and IPv6

Routing: Network-layer protocols



Services & Transport

Services: DNS, DHCP

Transport: TCP/UDP fundamentals



Security & Support

Access: Secure user configuration

Support: Help-desk troubleshooting

Network Architecture



Management Layer: Secure user access controls and structured troubleshooting/help-desk procedures.



Services Layer: DNS and DHCP servers providing name resolution and automatic address assignment.



Transport Layer: Protocols governing reliable and best-effort end-to-end host communication.



Network Layer: Routers using routing services and protocols for end-to-end connectivity.



Access Layer: Switches provide Ethernet connectivity; a wireless router/AP provides secure access.



End-Device Layer



Access Layer (Switch / AP)



Network Layer (Router)



Transport Layer



Services Layer (DNS/DHCP)

Program Journey (Methodology)

Phase 1

Networking Essentials

Foundational concepts: standards, protocols, Ethernet communication, IPv4/IPv6 addressing.

Phase 3

Career Path Capstone

Switched networks, wireless config, routing, troubleshooting & secure access — the verified credential.

Phase 2

Exploring with Packet Tracer

Hands-on simulation to build, configure and test small network topologies.

Each phase concluded with assessments; the capstone required demonstrating practical skills before the Cisco-verified credential was issued.

Key Skills & Demonstrated Competencies



Network Standards & Protocols: Explained the importance of standards and protocols governing network communications.



Switched Network Operation: Demonstrated how Ethernet and switching operate within a functioning switched network.



IPv4/IPv6 Addressing & Routing: Applied addressing schemes and configured routers for end-to-end connectivity.



Wireless & Secure Access: Configured an integrated wireless router/client and secured user access on the network.



Troubleshooting & Help Desk: Applied structured troubleshooting methodology and help-desk best practices to resolve DNS/DHCP-based connectivity issues.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Credential Title	Network Technician Career Path (Verified)
Issuing Organization	Cisco Networking Academy
Academic Partnership	IILM University, Greater Noida, U.P.
Supporting Courses	Networking Essentials; Exploring Networking with Cisco Packet Tracer
Issued On	June 16, 2026
Verified By	Lynn Bloomer, Director, Cisco Networking Academy

Certificate of Completion



Supporting Course Certificates



Thank You!



Ankit Kumar

Roll No: 2410031474



School of Computer Science &
Engineering

IILM University, Greater Noida, U.P.